

LAB 3: ENTERPRISE VIRTUALIZATION

CNIT24200-03

Group 3

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BUSINESS SCENARIO

The company WoodRock was a wood sales company originally housing three different domains under the company forest, with three department Organization Units (OU) being allocated to each domain. WoodRock started to prioritize Information Technology (IT) development as the company continued to grow. Virtualization was a great option to reduce energy and infrastructure costs, while being able to boost the company's scalability. WoodRock decided to implement Nutanix Community Edition, consisting of Nutanix Acropolis Hypervisor (AHV) and Nutanix Prism. The virtualizable environment made management much easier for the company, as Nutanix Prism Central offered management over several clusters. Using the hypervisor to Secure Shell (SSH) into the Controller Virtual Machine (CVM), WoodRock was able to configure the desired clusters. The hypervisor allowed hardware usage to be allocated to numerous machines, greatly increasing cost efficiency. WoodRock would incorporate two Virtual Machines (VM) housing Windows 2025 Servers, and each holding Active Directory Domain Services (AD DS). Server management looked seemingly the same for administrators except now access would be through the Nutanix VM instead of directly on the machine. The previous Active Directory domains, housing Windows 2022 Servers, would be demoted. The previous domain FSMO roles had been transferred to G03SRV01VM and G03SRV03VM, which were the new Windows 2025 Server VMs. WoodRock separated the VMs into two separate clusters, giving administrators an easy way to manage virtualization using Prism Central. Every VM had incorporated Nutanix VirtIO drivers, Nutanix Guest Tools (NGT), and remote desktop, further increasing accessibility and scalability to drive success. Virtual machines had memory overcommit enabled to grant efficient use of memory across all VMs, without requiring additional memory allocation.

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WoodRock saved millions in investing into Nutanix's bare-metal hypervisor, since purchasing additional hardware was no longer necessary. After the switch to Nutanix, energy costs fell and WoodRock saw exponential increases to revenue. Through virtualization, the company saw a secure and scalable way to support the enterprise's goals. In the figures below, the physical and logical models are displayed prior to the incorporation of enterprise virtualization through Nutanix.

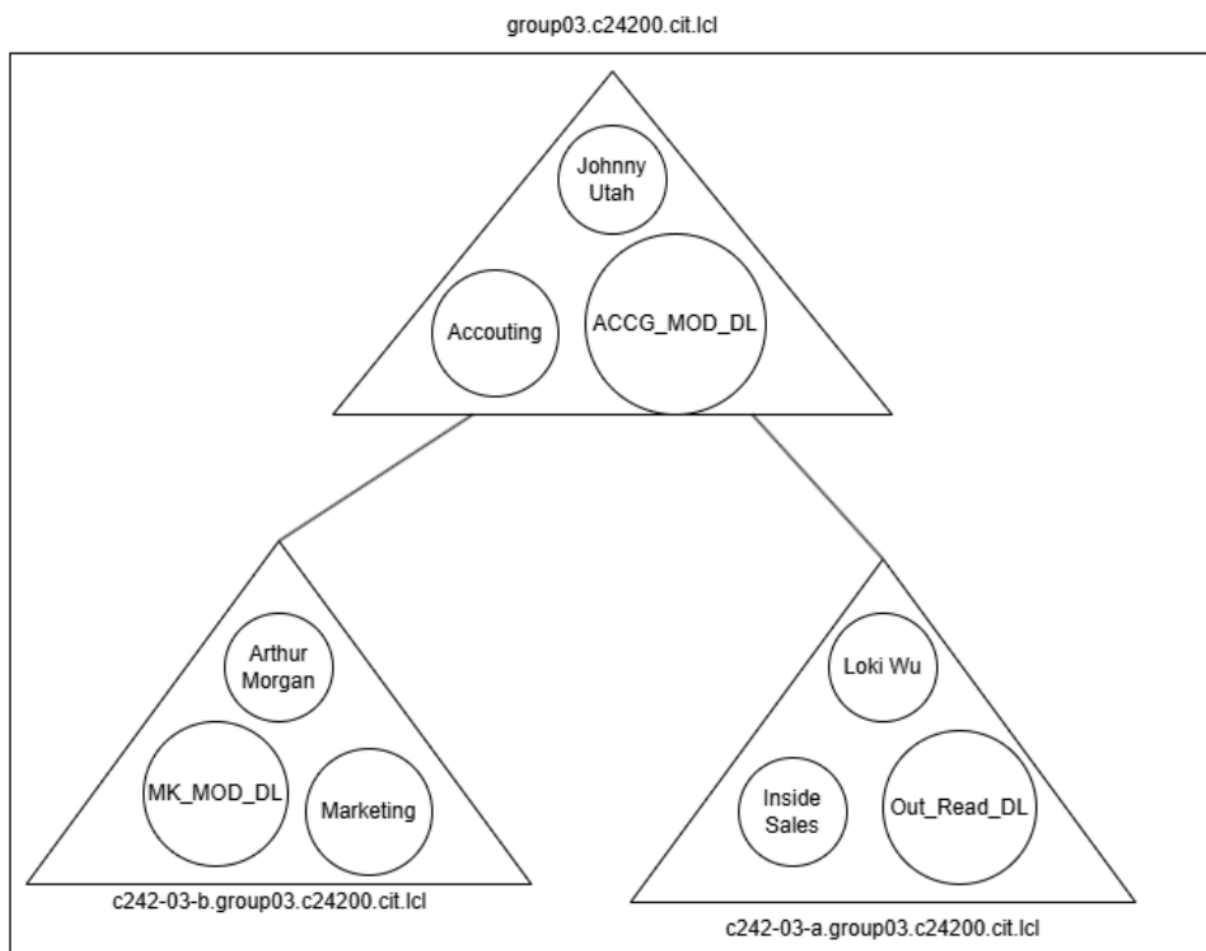


Figure 1: Diagram showing the logical model of the forest.

The figure above displayed three domains within the `group03.24200.cit.lcl` forest, being the parent, `c242-03-b`, and `c242-03-a`. Each domain possessed a security group, an Organizational Unit (OU), and a created user. The domains each held three OUs which represented a department of WoodRock.

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Figure 2: Diagram showing the physical model of the forest.

The physical model represented the environment in place for WoodRock to operate, which contained three total machines and their respective Internet Protocol (IP) addresses. All three servers (Windows 2022 Servers) pointed to tick.cit.lcl and tock.cit.lcl Network Time Protocol (NTP) servers, while the Computer Information Technology (CIT) network connected the machines to the k238 printer via the switch shown.

PROCEDURES

Each section contained a numbered list which showed the steps. Sections' headers, which were bolded and centered, showed the actions taken. The bibliography appeared at the end of the report, followed by Appendix A, which contained the problem statement. Additional supporting documentation was included in subsequent appendices. The followings were the formatting keys: **buttons** were bold, *options* were italicized, text entered into the computer was in Courier New, menu navigation was by the pipe symbol and italic words: *Start | Programs | MS Office | Word*.

G03SRV02 Domain Controller Demotion

The following section showed how G03SRV02 from an Active Directory (AD) Domain Controller (DC) to a non-Active Directory Domain was demoted. Demotion on G03SRV02 machine was needed for Nutanix Community Edition (CE) Acropolis Hypervisor (AHV) installation on the same machine. Unsuccessful DC demotion could cause AD domain conflicts. The following steps were documented on G03SRV02. Windows Server 2022 was expected to be booted on G03SRV02 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 prior to the first step.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager | Manage | Remove Roles and Features | Next* in *Before You Begin* window | **Next** in *Server Selection* window
4. Unchecked *Active Directory Domain Services*
5. Clicked *Remove Features | Demote this domain controller*
6. Checked *Last domain controller in the domain*
7. Clicked **Next** in *Credentials* window

8. Checked *Proceed with removal*
9. Clicked *Next* in *Warnings* window
10. Checked *Remove this DNS zone (this is the last DNS server that hosts the zone)* | *Remove application partitions* | *Remove DNS delegation*
11. Clicked **Change...**
12. Typed Administrator in *User name* and [c242-03-b password] in *Password*
13. Clicked **OK** in *Window Security* window | **Next** in *Removal Options* window
14. Typed [new c242-03-b password] in *Password* window and *Confirm password* window
15. Clicked **Next** in *New Administrator Password* | **Demote** in *Demotion* | **Close**

Bootable USB Configuration for Nutanix Optical Disk Image

The following section detailed how a bootable Universal Serial Bus (USB) was prepared with the use of Rufus software and Nutanix Optical Disk Image. A bootable USB was required for Nutanix CE AHV installation on the bare-metal server G03SRV02. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 prior to the first step.

1. Pressed **Windows + E**
2. Right-clicked *This PC* | *Map network drive...*
3. Typed \\rtfm.cit.lcl in *Folder*
4. Clicked *Browse...*

5. Typed CIT\[CIT student username] in *User name* and [CIT student password] in *Password*
6. Clicked *OK* in *Windows Security* window
7. Expanded *rtfm.cit.lcl* | *Pub* | *CNIT24200*
8. Clicked *OK* in *Browse For Folder* window | *Finish*
9. Right-clicked *rufus-4.7* | *Open*
10. Clicked *Run* | *No* in *Rufus update policy* window | *SELECT* | *CNIT24200*
(\\rtfm.cit.lcl\Pub) | *phoenix.x86_64-fnd_5.6.1_patch-aos_6.8.1_ga* | *Opened* | *START* |
OK in *Rufus* | *OK* in *Multiple partitions detected*

G03SRV02 Partitions Deletion

The following section detailed the deletion of all existing disk partitions on G03SRV02. Clearing partitions prepared the environment for the Nutanix CE AHV installation. The following steps were documented on G03SRV02. Windows Server 2022 was expected to be booted on G03SRV02 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 and “UEFI Lexar USB Flash” was expected to be plugged in G03SRV02 before the first step was started prior to the first step.

1. Pressed **Windows**
2. Clicked *Power* | *Restart*
3. Pressed **F12**
4. Clicked *UEFI Lexar USB Flash Drive 9707422273028248* in *One-Time Boot Settings*

5. Pressed **Enter** | **Enter** on
en-us_windows_server_2022_updated_july_2023_x64_dvd_541692c3 | **Enter** on *Boot in normal mode*
6. Clicked **Next** in *Microsoft Server Operating System Setup* window | *Install now* | *I don't have a product key* | *Windows Server 2022 Standard (Desktop Experience)* | *Next* in
Select the operating system you want to install
7. Checked *I accept the Microsoft Software License Terms. If an organization is licensing it, I am authorized to bind the organization*
8. Clicked *Next* in *Applicable notices and license terms* | *Custom: Install Microsoft Server Operating System only (advanced)*
9. Clicked [Drive 0/1/2 partitions] | *Delete* | *OK* in *Microsoft Server Operating System Setup* window
10. Repeated step 9 until all [Drive 0/1/2 Unallocated Space]
11. Closed *Microsoft Server Operating System Setup* window | *Yes* in *Microsoft Server Operating System Setup* window
12. Repeated step 11

Nutanix Community Edition Acropolis Hypervisor Installation on G03SRV02

The following section detailed the installation of Nutanix CE AHV on G03SRV02. The hypervisor was configured with a static Internet Protocol (IP) address, subnet mask, gateway, and Controller Virtual Machine (CVM) IP address, which were necessary for G03SRV02 to be integrated into the network. The following steps were documented on G03SRV02. G03SRV02 was expected to be powered off and UEFI Lexar USB Flash Drive was expected to be plugged in before the first step was started. **Refer to Appendix B.**

1. Powered *G03SRV02*
2. Pressed **F12**
3. Selected *UEFI Lexar USB Flash Drive 6164134633453092* in *One-Time Boot Settings*
4. Pressed **Enter** in *nvme0n1*
5. Typed *44.38.3.3* in *Host IP Address*
6. Typed *255.255.255.0* in *Subnet Mask*
7. Typed *44.38.3.1* in *Gateway*
8. Typed *44.38.3.10* in *CVM IP Address*
9. Selected *Next Page*
10. Checked *I accept the end user license agreement.*
11. Selected **Start**

Nutanix Community Edition Acropolis Hypervisor Configuration on G03SRV02

The following section detailed the post-installation configuration of the Nutanix cluster on G03SRV02, which was later renamed into G03NUTXSRV02. Steps included cluster creation, cluster name configuration, Domain Name System (DNS) configuration, and Network Time Protocol (NTP) server configuration. Time synchronization and proper network settings were important. The following steps were documented on G03SRV02. The G03SRV02 was expected to be powered on before the first step was started.

1. Typed `root` in *NTNX-c8d32b3b-A login*
2. Typed `nutanix/4u` in *Password*
3. Typed `ssh nutanix@192.168.5.2` in *[root@NTNX-c8d32b3b-A ~]#*
4. Typed `yes` in *Are you sure you want to continue connecting*
5. Typed `nutanix/4u` in *nutanix@192.168.5.2's password*
6. Typed `cluster -s 44.38.3.10 --redundancy_factor=1 create` in *[root@NTNX-c8d32b3b-A ~]#*
7. Typed `ncli cluster edit-params new-name=G03NUTXSRV02` in *nutanix@NTX-c8d32b3b-A-CVM:44.38.3.10:~\$*
8. Typed `ncli cluster add-to-name-servers servers=44.2.1.44` in *nutanix@NTX-c8d32b3b-A-CVM:44.38.3.10:~\$*
9. Typed `ncli cluster add-to-ntp-servers servers=tick.cit.lcl` in *nutanix@NTX-c8d32b3b-A-CVM:44.38.3.10:~\$*
10. Typed `ncli cluster add-to-ntp-servers servers=tock.cit.lcl` in *nutanix@NTX-c8d32b3b-A-CVM:44.38.3.10:~\$*

Signing Into G03NUTXSRV02 Prism Element Web Page

The following procedures were executed from the desktop of any machine, considering the machine was connected to the Computer Information Technology (CIT) network or the CIT Virtual Private Network (VPN). Each task was completed by following the given prompts and entering the Nutanix Prism Element admin credentials, where the password would be changed. The procedures were executed to manage G03NUTXSRV02 cluster within Nutanix Prism Element, and provide reference for future Prism Element procedure sections. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 prior to the first step.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.10:9440` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked **Advanced** on Google security page | *Continue to 44.38.3.10 (unsafe)*
5. Typed `admin` in *username*
6. Typed `nutanix/4u` in *password*
7. Typed `[new secure password]` in *password*
8. Typed `[new secure password]` in *re-type password*
9. Repeated step 5
10. Repeated step 7

Creating NEXT Account for Prism Element

The section detailed the registration of a Nutanix NEXT community account, which enabled functionality and support for the virtualization platform. Establishment of the account was necessary to license the Community Edition software and access the Nutanix environment. [sign up field] referred to the field the user had to type in, which could be First name, Last name, Company name, Job title, Phone number, Country, Email, and Password. [sign up field input] referred to the input which was uniquely assigned to the [sign up field]. The following steps were documented on G03SRV01. Windows Server 2022 was expected to be booted on G03SRV01 machine before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step.

1. Clicked *Create account* in *NEXT Credentials* window
2. Clicked *Sign up now* in *Log in with your Email* window
3. Typed [sign up field input] in [sign up field]
4. Repeated step 3 for each [sign up field]
5. Clicked **Submit**
6. Pressed **Ctrl + T**
7. Typed outlook.office365.com/mail/junkemail
8. Pressed **Enter**
9. Clicked *MyNutanix | Confirm email | Visit Link* in *Junk Email Security* window |

Continue

10. Typed [student secure password] in *Password*
11. Clicked **Log in** | **Next** in *Introducing Workspaces* window | **Next** in *Switch between multiple workspaces* window | **Next** in *We've created a personal workspace for you*

window | **Next** in *Create & manage up to 25 workspaces* window | **Close** in *Switch workspaces from the account menu* window | **Launch** in *Community Edition* window

Prism Element Logging

The section detailed the authentication process to access the single cluster management dashboard, which was Nutanix Prism Element (PE). Microsoft Edge browser was used to perform login and administrative tasks on Nutanix PE. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step.

1. Typed `https://44.38.3.10:9440/console` in *Microsoft Edge Searchbar*
2. Pressed **Enter**
3. Typed `admin` in username
4. Typed `[new secure password]` in password
5. Pressed **Enter**

iSCSI Data Service IP and Virtual IP Configuration on G03NUTXSRV02

The procedures below were executed within the Nutanix Prism Element console for the G03NUTXSRV02 cluster. IP addresses were generated to configure both the Internet Small Computer System Interface (iSCSI) and Virtual IPs for the cluster. The steps were necessary for the cluster to become discoverable by Nutanix volumes, and so the cluster could be referenced by the CVM. The following steps were documented on G03SRV03. G03SRV02 machine and G03SRV03 machine were expected to be powered on before the first step was started. Windows Server 2022 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 prior to the first step. **Refer to Appendix B.**

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Pressed **Windows + T**
3. Typed `https://44.38.3.10:9440/console/` in *Microsoft Edge Searchbar*
4. Pressed **Enter**
5. Clicked *Home | Settings | Cluster Details*
6. Typed `44.38.3.42` in *iSCSI Data Services IP*
7. Typed `44.38.3.41` in *Virtual IP*
8. Clicked **Save**

Prism Central Deployment on G03NUTXSRV02

The section detailed the deployment of Nutanix Prism Central (PC) on the G03NUTXSRV02 cluster. PC provided a centralized management interface for multiple Nutanix clusters and was configured with a dedicated VM, IP address, and network subnet. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section). **Refer to Appendix B.**

1. Clicked *Home | Settings | Cluster Details | Prism Central Registration | Register or create new | **Deploy*** under I want to deploy a new Prism Central instance | *pc.2024.3.1.9 | **Next*** in *PC Version* window | *Small (S) | **Next*** in *PC Size and Scale* window | *Select or create a Subnet* dropdown | *Create Network*
2. Typed G03NUTXSRV02 in *Subnet Name*
3. Entered 0 in *VLAN ID*
4. Clicked **Save**
5. Typed 255.255.255.0 in *Subnet Mask* field
6. Typed 44.38.3.1 in *Gateway IP*
7. Typed 44.2.1.44, 44.2.1.45 in *DNS Address(es)* field
8. Typed 44.2.1.44, 44.2.1.45 in *NTP Address(es)* field
9. Clicked *SelfServiceContainer* dropdown | *default-container-66090046333395*
10. Typed G03NUTXSRV02 in *VM Name*
11. Typed 44.38.3.12 in *IP Address*
12. Clicked **Next** in *Configuration | Private Network [default] | G03NUTXSRV02*

13. Entered 44.38.3.13 – 44.38.3.17 within *IP Address Range (5 IPs needed)* field
14. Clicked **Deploy**

Storage Container Creation on G03NUTXSRV02

The section detailed the creation of a new storage container named “G03VM” within the Prism Element interface. Storage container creation was important for organizing virtual machine data, which provided the necessary environment where the new virtual disks would reside. The following steps were documented on G03SRV03. G03SRV02 machine and G03SRV03 machine were expected to be powered on before the first step was started. Windows Server 2022 was expected to be booted on G03SRV03 before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step. **Refer to Appendix B.**

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.10:9440/console/` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked *Home* dropdown | *Storage* | *Table* | + *Storage Container*
5. Typed G03VM in *Name*
6. Clicked **Save**

Disabling Internet Protocol Version 6 on G03NUTXSRV02

The section detailed the command-line procedures performed on the CVM, which disabled IP version 6 (IPv6) and modified `sysctl.conf`. To prevent potential communication conflicts, IPv6 was disabled at the hypervisor level. The following steps were documented on G03NUTXSRV02. G03NUTXSRV02 was expected to be powered on before the first step was started

1. Typed `root` in *NTNX-c8d32b3b-A login*
2. Typed `nutanix/4u` in *Password*
3. Typed `ssh nutanix@192.168.5.2` in *[root@NTNX-c8d32b3b-A ~]#*
4. Typed `[new secure password]` in *root@NTNX-c8d32b3b-A ~'s password*
5. Typed `sudo sysctl -w net.ipv6.conf.all.disable_ipv6=1` in *[nutanix@NTNX-c8d32b3b-A ~CVM]#*
6. Pressed **Enter**
7. Typed `echo -e "net.ipv6.conf.all.disable_ipv6 = 1\nnet.ipv6.conf.default.disable_ipv6 = 1\nnet.ipv6.conf.lo.disable_ipv6 = 1" | sudo tee -a /etc/sysctl.conf > /dev/null` in *nutanix@NTNX-c8d32b3b-A ~CVM*
8. Typed `sudo sysctl -p` in *nutanix@NTNX-c8d32b3b-A ~CVM*
9. Typed `sudo reboot` in *nutanix@NTNX-c8d32b3b-A ~CVM*
10. Typed `manage_ipv6 unconfigure` in *nutanix@NTNX-c8d32b3b-A ~CVM:44.38.3.10*
11. Pressed **Enter**

12. Typed `manage_ipv6 unconfigure` in *nutanix@NTNX-c8d32b3b-A*

~CVM:44.38.3.10

13. Pressed **Enter**

14. Typed `rolling_restart cvms` in *nutanix@NTNX-c8d32b3b-A ~CVM*

15. Pressed **Enter**

Connecting Cluster Prism Element on G03NUTXSRV02 to Nutanix Prism Central

The section detailed the registration process where the G03NUTXSRV02 cluster was linked to the Nutanix PC instance at IP 44.38.3.12, which made cross-cluster resource management possible. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Clicked *Home | Settings | Cluster Details | Prism Central Registration | Register or create new | **Connect***
2. Typed `44.38.3.12` in *IP Address*
3. Typed `9440` in *Port*
4. Typed `admin` in *Username*
5. Typed `[new secure password]` in *Password*
6. Clicked **Connect**

Password Reset

The section detailed the reset of password for default Nutanix PC administrative account. Password reset was necessary to protect the Nutanix environment from unauthorized access. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Pressed **Windows + T**
3. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
4. Pressed **Enter**
5. Typed `admin` in *Username*
6. Typed `Nutanix/4u` in *Password*
7. Typed [new secure password] in *New Password*
8. Typed [new secure password] in *Retype New Password*
9. Typed `Group03` in *Name*
10. Typed `Purdue` in *Company*
11. Typed `Students` in *Job Title*
12. Checked *I have read and agree to the terms and conditions*
13. Clicked **Accept | Continue | Skip** in *What's New?* Window

Prism Central Logging

The section detailed the authentication process to access the centralized management dashboard for the Nutanix environment, which was Nutanix PC. Microsoft Edge browser was used to perform login and administrative tasks on Nutanix PC. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Pressed **Windows + T**
3. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
4. Pressed **Enter**
5. Typed `admin` in *Username*
6. Typed `[new secure password]` in *Password*

Image Configuration

The section detailed the process of mapping a network drive from the university repository to G03SRV03 machine, which was necessary to import optical disc images for Windows 2025, Windows 11, and Nutanix VirtIO drivers. Centralizing Optical Disk Images allowed rapid and consistent VM deployment across the virtualized environment. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Clicked *Compute* dropdown | *Images* | **Add Image** | **+ Add File**
2. Right-clicked *This PC* | *Map network drive*
3. Typed `\\rtfm.cit.lcl`
4. Clicked **Browse...** | *rtfm.cit.lcl*
5. Typed `CIT\[school account username]` in *User Name*
6. Typed `[school account password]` in *Password*
7. Clicked *rtfm.cit.lcl* | *Pub* | *CNIT24200* | **OK** in *Browse For Folder* window | **Finish** | *en-us_windows_server_2025_x64_dvd_b7ec10f3.iso* in *Open* window | **Open** in *Open window*
8. Clicked **Next** in *Select Image* | **Save** in *Select Location*
9. Clicked **Add Image** | **+ Add File** | *Nutanix-VirtIO-1.2.5.iso* in *Open* window | **Open** | **Next** in *Select Image* | **Save** in *Select Location*
10. Clicked **Add Image** | **+ Add File** | *en-us_windows_11_consumer_editions_version_24h2_x64_dvd_1d5fcad3.iso* in *Open window* | **Open**

11. Typed `Windows11` in *Name*
12. Clicked **Next** in *Select Image* | **Save** in *Select Location*

G03WIN2025TEMP Deployment

The section detailed the creation of a Windows Server 2025 VM, which was then configured with the necessary virtual hardware like virtual Central Processing Unit (vCPU), memory, VirtIO driver, and Serial Advanced Technology Attachment (SATA) disks. The VM would serve as a standardized baseline for all subsequent Windows Server 2025 VM instances. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section). **Refer to Appendix B.**

1. Clicked *Compute* on dashboard page | *VMs* under compute dropdown | **Create VM**
2. Typed `G03WIN2025TEMP` in *Name*
3. Checked *Enable Memory Overcommit*
4. Typed `2` in *Cores Per CPU*
5. Typed `8` in *Memory*
6. Clicked **Next** in *Configuration* | *Attach Disk* | *Type* dropdown | *CD-ROM* | *Operation* dropdown | *Clone from Image* | *Image* dropdown | *en-us_windows_server_2025_x64_dvd_b7ec10f3.iso* | **Save**
7. Selected *Attach Disk* | *Type* dropdown | *CD-ROM* | *Operation* dropdown | *Clone from Image* | *Image* dropdown | *Nutanix-VirtIO-1.2.5.iso* | **Save**
8. Clicked *Attach Disk* | *Type* dropdown | *Disk* | *Operation* dropdown | *Allocate on Storage Container* | *Storage Container* dropdown | *G03VM* | *Bus Type* dropdown | *SATA*

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9. Typed 250 in *Capacity*
10. Clicked **Save** | **Attach to Subnet** | **Save**
11. Checked *Secure Boot* | *Attach vTPM*
12. Clicked **Next** in *Resources* | *Timezone* dropdown | (*UTC - 05:00*)
America/Indiana/Indianapolis | **Next** in *Management* | **Create VM**

Windows Server 2025 Installation on G03WIN2025TEMP

The section detailed the installation of Windows Server 2025 (Desktop Experience) on the template VM called G03WIN2025TEMP. The process included booting from the attached ISO, configuring language and keyboard settings, and establishing the initial administrative credentials for the virtual server. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Clicked *Compute* dropdown | *VMs* | *G03WIN2025TEMP* | *More* dropdown on G03WIN2025TEMP page | *Power On* | **Launch Console**
2. Pressed **Enter** in *Press any key to boot from CD or DVD*
3. Selected *This selection will take you to the Boot Manager* | **Enter** | **Enter** in *Boot Manager (on UEFI QEMU DVD-ROM QM00001)* | **Enter** in *Press any key to boot from CD or DVD* | **Enter** in *Windows Boot Manager (windows setup button)*
4. Clicked **Next** in *Select language settings* window | **Next** in *Select keyboard settings* |
5. Checked *I agree everything will be deleted including files, apps, and settings*
6. Clicked **Next** in *Select setup option* | *I don't have a product key* | *Windows Server 2025 Standard (Desktop Experience)* | **Next** in *Select Image* | **Accept** in *Applicable notices and license terms* | **Next** in *Select location to install Windows Server* | **Install**
7. Selected *Do this later*
8. Typed [new secure password] in *Password*
9. Inputted [new secure password] in *Reenter Password*
10. Clicked **Finish** | **Send CtrlAltDel (button icon)**
11. Typed [new secure password] in *Password*

G03WIN2025TEMP Opening

The section detailed the initialization of a remote console session to interact with the guest operating system of the template machine, which was necessary to perform internal VM system configurations. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step.

1. Pressed **Ctrl + T**
2. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked *Compute* dropdown | *VMs* | *G03WIN2025TEMP* | *Launch Console* | **Send CtrlAltDel**
5. Typed [secure password] in *Password*
6. Pressed **Enter**

VirtIO, Sysprep, and Remote Desktop Configurations on G03WIN2025TEMP

The section detailed the installation of Nutanix VirtIO drivers, execution of Sysprep, and configuration of Remote Desktop on G03WIN2025TEMP. VirtIO driver optimized VM performance on AHV. System generalized the Operating System (OS) image to be deployed as a reusable template. Remote Desktop allowed seamless remote interaction with the VM. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the G03WIN2025TEMP desktop was expected to be launched (see G03WIN2025TEMP Opening section).

1. Clicked **Accept** | *Local Server* | *Disabled in Remote Desktop* | *Allow remote connections to this computer* | **OK** in *Remote Desktop Connection* | **OK** in *System Properties*
2. Clicked *WIN-UBRASDUN6T5* computer name | **Change...**
3. Typed *G03WIN2025TEMP* under computer name
4. Clicked **OK** in *Computer Name/Domain Changes* window | **OK** in *You must restart your computer to apply these changes* window | **Close** in *System Properties* | **Restart Now**
5. Clicked ***Send CtrlAltDel (button icon)*** |
6. Typed [secure password] in *Password*
7. Clicked *File Explorer* in *Windows Taskbar* | *CD Drive (E:) Nutanix VirtIO 1.2.5* |
8. Double-clicked *Nutanix VirtIO 1.2.5-x64*
9. Checked *I accept the terms in the License Agreement*
10. Clicked **Install** | **Finish** | **Unmount ISO** in *G03WIN2025TEMP - Profile 1 - Microsoft Edge* window | *Select disk* dropdown | *SATA. 1 - Nutanix-VirtIO-1.2.5.iso* | **Unmount**

11. Clicked **Unmount ISO** in *G03WIN2025TEMP - Profile 1 - Microsoft Edge* window |
Select disk dropdown | *SATA. 0 -*
en-us_windows_server_2022_updated_july_2023_x64_dvd_541692c3.iso | **Unmount**
12. Typed `cd C:\Windows\system32\sysprep; .\sysprep /generalize`
`/shutdown /oobe`
13. Pressed **Enter**
14. Clicked *Close* in *G03WIN2025TEMP - Profile 1 - Microsoft Edge* window

Mounting Nutanix Guest Tools on G03WIN2025TEMP

The section detailed the process of enabling and mounting the Nutanix Guest Tools (NGT) installation media to the G03WIN2025TEMP. NGT provided better management capabilities like self-service restore (SSR) and snapshots application. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 before the first step was documented; likewise, the Prism Element browser console was expected to be launched (see Prism Element Logging section).

1. Clicked *Home* dropdown | *VM*
2. Right-clicked *G03WIN2025TEMP*
3. Clicked *Manage Guest Tools*
4. Checked *Enable Nutanix Guest Tools* | *Mount Nutanix Guest Tools*
5. Clicked **Submit** in *Manage VM Guest Tools*

G03WIN2025TEMP Snapshot and Template Creation

The section detailed the creation of a recovery point snapshot and a VM template from G03WIN2025TEMP. The process included the creation of a recovery point named “WIN2025Config”. The template allowed consistent and fast deployment of pre-configured virtual machines without reinstalling and reconfiguring the operating system from scratch. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Pressed **Windows + T**
2. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Typed `admin` in *Username*
5. Typed `[new secure password]` in *Password*
6. Clicked *Compute* dropdown | *VMs* | *G03WIN2025TEMP* | *More* dropdown | *Create Recovery Point*
7. Typed `WIN2025Config`
8. Clicked **Create** in *Create Recovery Point* | **Create VM Template**
9. Clicked **Next** in *Create Template from VM* | **Save**

G03SRV03VM Deployment

The section detailed the deployment of G03SRV03VM from the G03WIN2025TEMP template. Deploying from a template ensured a standardized configuration across VMs in the environment and reduced setup time. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section). **Refer to Appendix B.**

1. Clicked *Compute* dropdown | *VMs* | **Create VM from Template** | *VM Template* dropdown | *G03WIN2025TEMP* | **Begin**
2. Typed G03SRV03VM in *Name*
3. Clicked **Next** in *Deploy VM from Template* | **Deploy** | *G03SRV03VM* | *More* dropdown | *Power On* | **Launch Console** | *Send CtrlAltDel*
4. Typed [secure password] in *Password*
5. Pressed **Enter**

G03SRV03VM Opening

The section detailed the procedure for launching the VM console from the Nutanix Compute dashboard and authenticating with the established administrative credentials.

Consistent access to the VM console was necessary for tasks which involved G03SRV03VM.

The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step

1. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
2. Pressed **Enter**
3. Clicked *Compute* dropdown | *VMs* | *G03SRV03VM* | *Launch Console* | ***Send CtrlAltDel (button icon)***
4. Typed [secure password] in *Password*
5. Pressed **Enter**

IP Address Configuration on G03SRV03VM

The section detailed the process of disabling IPv6 and assigning static IP version 4 (IPv4) address (44.38.2.20) and Domain Name System (DNS) settings to G03SRV03VM. Proper IP configuration allowed the new server to communicate with existing domain controllers and services. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV03VM desktop was expected to be launched (see G03SRV03VM Opening section).

Refer to Appendix B.

1. Clicked *Local Server* | *IPv4 address assigned by DHCP, IPv6 enabled in Ethernet*
2. Double-clicked *Ethernet*
3. Clicked **Properties**
4. Unchecked *Internet Protocol Version 6 (TCP/IPv6)*
5. Double-clicked *Internet Protocol Version 4 (TCP/IPv4)*
6. Checked *Use the following IP address* and *Use the following DNS server addresses*
7. Typed 44.38.3.20 in *IP address*
8. Typed 255.255.255.0 in *Subnet mask*
9. Typed 44.38.3.1 in *Default gateway*
10. Typed 44.38.3.4 in *Preferred DNS server*
11. Clicked **OK** in *Internet Protocol Version 4 (TCP/IPv4) Properties* | **OK** *Ether Properties* | **Close** in *Ethernet Status* | **Close in** *Network Connections*

Domain Configuration on G03SRV03VM

The section detailed the process of enrolling G03SRV03VM from a workgroup to the c242-03-a.group03.c24200.cit.lcl domain. The process allowed the server to participate in domain-based security, Group Policy, and centralized management. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV03VM desktop was expected to be launched (see G03SRV03VM Opening section). **Refer to Appendix B.**

1. Clicked *Local Server* | *G03SRV03VM* | **Change...**
2. Typed G03SRV03VM in *Computer name*
3. Checked *Domain*
4. Typed c242-03-a.group03.c24200.cit.lcl
5. Clicked **OK** in *Computer Domain Name Changes*
6. Typed *Administrator* in *Username*
7. Typed [Domain Administrator password] in *Password*
8. Clicked **OK** in *Windows Security* | **OK** in *Computer Domain/Name Changes*
9. Clicked **Close** in *System Properties* | **Restart Now** in *Microsoft Windows* window

Adding C242-03-A\Administrator into Schema Admins and Enterprise Admins Groups

The section detailed the process of adding the Active Directory (AD) domain administrator account to the Schema Admins and Enterprise Admins security groups using AD Users and Computers, which granted the necessary permission to promote G03SRV03VM from a normal AD domain computer user to a DC. The following steps were documented on G03SRV01. Windows Server 2022 was expected to be booted on G03SRV01 before the first step was documented; likewise, the G03SRV03VM desktop was expected to be launched (see G03SRV03VM Opening section).

1. Clicked *Server Manager* in Windows Taskbar | *Tools* | *Active Directory Users and Computers* | *group03.c24200.cit.lcl* | *Users*
2. Right-clicked *Enterprise Admins*
3. Clicked *Properties* | *Members* | **Add...** | **Advanced..** | **Locations...** | *group03.c24200.cit.lcl* | *c242-03-a.group03.c24200.cit.lcl* | *Users* | **OK** in *Locations* | **Find Now** | *Administrator*
4. Clicked **OK** in *Select Users, Contacts, Computers, Service Accounts, or Groups* | **OK** in *Select Users, Contacts, Computers, Service Accounts, or Groups* **OK** in *Enterprise Admins Properties*
5. Right-clicked *Schema Admins*
6. Repeated steps 3-4
7. Closed *Active Directory Users and Computers* window

Promoting G03SRV03VM to Domain Controller

The section detailed the promotion of G03SRV03VM to an AD DC within c242-03-a.group03.c24200.cit.lcl domain. The objective was completed through navigation within the features of Active Directory server manager, and in ensuring the parent DC administrator credentials used during installation, so the prerequisite checks can run successfully. Domain controller abilities were necessary on server G03SRV03VM, so Flexible Single Master Operations (FSMO) roles could later be transferred to the DC. The following steps were documented on G03SRV03. Windows Server 2025 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV03VM desktop was expected to be launched (see G03SRV03VM Opening section).

1. Clicked *Manage | Add Roles and Features | Next > in Before You Begin | Next > in Installation type | Next > in Server Selection*
2. Checked *Active Directory Domain Service | DNS Server*
3. Clicked **Add Features** | **Next >** in *Server Roles* | **Next >** in *Features* | **Next >** in *AD DS | Install in Confirmation | Close in Results | (Notification icon) in Server Manager | Promote this server to a domain controller | Change...*
4. Typed `Group03\Administrator` in *User name*
5. Typed `[Administrator password from lab 2]` in *Password*
6. Clicked **OK** in *Windows Security | Select... | c242-03-a.group03.c24200.cit.lcl | OK in Select a domain from the forest | Next > in Deployment Configuration*
7. Typed `[New DSRM password]` in *Password*
8. Typed `[New DSRM password]` in *Confirm password*

9. Clicked **Next** > in *Domain Controller Options* | **Next** > in *DNS Options* | **Next** > in *Additional Options* | **Next** > in *Paths* | **Change...**
10. Typed Administrator in *User name*
11. Typed [Administrator password from lab 2] in *Password*
12. Clicked **OK** in *Windows Security* | **Next** in *Preparation Options* | **Next** in *Review Options* | **Install** in *Prerequisites checks*

FSMO Roles Transfer on G03SRV03

The section detailed the transfer of FSMO roles from G03SRV03 server to G03SRV03VM server. FSMO roles transferred allowed the new virtual DC to take over AD operations, which included Primary Domain Controller (PDC) Emulator, Relative Identifier (RID) Master, and Infrastructure Master functions. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 prior to the first step.

1. Clicked *Server Manager* in Windows Searchbar | *Tools* | *Active Directory Users and Computers*
2. Right-clicked *Active Directory Users and Computers*
3. Clicked *Change Domain Controller...* | *G03SRV03VM.c242-03-a.group03.c24200.cit.lcl* | **OK** in *Change Directory Server*
4. Right-clicked *c242-03-a.group03.c24200.cit.lcl*
5. Clicked *Operations Masters...* | **Change...** | **Yes** in *Active Directory Domain Services* | **OK** in *Active Directory Domain Services* |

6. Clicked *PDC* | **Change...** | **Yes** in *Active Directory Domain Services* | **OK** in *Active Directory Domain Services*
7. Clicked *Infrastructure* | **Change...** | **Yes** in *Active Directory Domain Services* | **OK** in *Active Directory Domain Services*
8. Clicked **Close** in *Operations Masters*
9. Pressed **Windows + R**
10. Typed `powershell` in *Open*
11. Clicked **OK** in *Run*
12. Typed `repadmin /replsummary; repadmin /showrepl` in *PS*
C:\Users\Administrator
13. Pressed **Enter**

G03WIN11TEMP Deployment

The section detailed the creation of the G03WIN11TEMP VM within Nutanix PC, which was then configured with the necessary virtual hardware like vCPU, memory, SATA disks... The VM would serve as a standardized baseline for all subsequent Windows 11 VM instances. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section). **Refer to Appendix B.**

1. Clicked *Compute* on dashboard page | *VMs* under compute dropdown | **Create VM**
2. Typed G03WIN11TEMP in *Name*
3. Checked *Enable Memory Overcommit*
4. Typed 2 in *Cores Per CPU*

5. Typed 4 in *Memory*
6. Clicked **Next** in *Configuration* | *Attach Disk* | *Type* dropdown | *CD-ROM* | *Operation*
7. dropdown | *Clone from Image* | *Image* dropdown |
us_windows_11_consumer_editions_version_24h2_x64_dvd_1d5fcad3.iso | **Save**
8. Clicked *Attach Disk* | *Type* dropdown | *Disk* | *Operation* dropdown | *Allocate on Storage*
Container | *Storage Container* dropdown | *G03VM* | *Bus Type* dropdown | *SATA*
9. Typed 60 in *Capacity*
10. Clicked **Save** | **Attach to Subnet** | **Save**
11. Checked *Secure Boot* | *Attach vTPM*
12. Clicked **Next** in *Resources* | *Timezone* dropdown | (*UTC - 05:00*)
America/Indiana/Indianapolis | **Next** in *Management* | **Create VM**

Windows 11 Installation on G03WIN11TEMP

The section detailed the installation of Windows 11 (Education) on the template VM called G03WIN11TEMP. The process included booting from the attached ISO, configuring language and keyboard settings, and establishing the initial administrative credentials for the virtual server. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section). The previous section was expected to have been properly executed.

1. Clicked *Compute* dropdown | *VMs* | *G03WIN2025TEMP* | *More* dropdown on *G03WIN11TEMP* page | *Power On* | **Launch Console**
2. Pressed **Enter** in *Press any key to boot from CD or DVD* | **Enter** in *Boot Manager (on UEFI QEMU DVD-ROM QM00001)* | **Enter** in *Press any key to boot from CD or DVD* | **Enter** in *Windows Boot Manager (windows setup button)*
3. Clicked **Next** in *Select language settings* window | **Next** in *Select keyboard settings*
4. Checked *I agree everything will be deleted including files, apps, and settings*
5. Clicked **Next** in *Select setup option* | *I don't have a product key* | *Windows 11 Education* | **Next** in *Select Image* | **Accept** in *Applicable notices and license terms* | **Next** in *Select location to install Windows Server* | **Install** | **Yes** in *Is this the right country or region*
6. Selected **Yes** in *Is this the right keyboard layout or input method* | **Skip** in *Want to add a second keyboard layout* | *I don't have internet* in *Let's connect you to a network*
7. Typed Admin in *Enter your name*
8. Clicked **Next** in *Who's going to use this device*
9. Typed [new admin secure password] in *Enter a password*

10. Inputted [new admin secure password] in *Password confirmation*
11. Typed [answer] in *Security question (1 of 3)*
12. Clicked **Next** in *Now add security questions*
13. Repeated steps 11-12 in *Security question (2 of 3)*
14. Repeated steps 11-12 in *Security question (3 of 3)*
15. Clicked **Accept** in *Choosing privacy settings for your device*

G03WIN11TEMP Opening

The section detailed the procedure for launching the VM console from the Nutanix Compute dashboard and authenticating with the established administrative credentials. Consistent access to the VM console was necessary for tasks which involved G03WIN11TEMP. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step

1. Pressed **Ctrl + T** in *Microsoft Edge* browser
2. Typed https://44.38.3.12:9440/ui/iam/login in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked *Compute* dropdown | *VMs* | *G03WIN11TEMP* | *Launch Console* | **Send**
CtrlAltDel
5. Typed [secure password] in *Password*
6. Pressed **Enter**

Remote Desktop and Sysprep Configurations on G03WIN11TEMP

The section detailed the execution of Sysprep and configuration of Remote Desktop on G03WIN11TEMP. System generalized the Operating System (OS) image to be deployed as a reusable template. Remote Desktop allowed seamless remote interaction with the VM. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the G03WIN11TEMP desktop was expected to be launched (see G03WIN11TEMP Opening section).

1. Right-clicked **Start**
2. Clicked *Settings* | *System* | *Remote Desktop* | *On* | **Confirm** in *Remote Desktop Settings* window
3. Clicked **Install** | **Finish** | **Unmount ISO** in *G03WIN2025TEMP - Profile 1 - Microsoft Edge* window | *Select disk* dropdown | *SATA. 0 - Windows11.iso* | **Unmount**
4. Right-clicked **Starts**
5. Clicked *Settings* | *System* | *Rename*
6. Typed G03WIN11TEMP in *Current PC name*
7. Clicked **Next** in *Rename your PC* | **Restart Later**
8. Clicked *Powershell* in Windows Taskbar
9. Typed `cd C:\Windows\system32\sysprep; .\sysprep /generalize /shutdown /oobe`
10. Pressed **Enter**
11. Clicked *Close* in *G03WIN2025TEMP - Profile 1 - Microsoft Edge* window

Mounting Nutanix Guest Tools on G03WIN11TEMP

The section detailed the process of enabling and mounting the NGT installation media to the G03WIN11TEMP. NGT provided better management capabilities like self-service restore (SSR) and snapshots application. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 before the first step was documented; likewise, the Prism Element browser console was expected to be launched (see Prism Element Logging section).

1. Clicked *Home* dropdown | *VM*
2. Right-clicked *G03WIN2025TEMP*
3. Clicked *Manage Guest Tools*
4. Checked *Enable Nutanix Guest Tools* | *Mount Nutanix Guest Tools*
5. Clicked **Submit** in *Manage VM Guest Tools*

G03WIN11TEMP Snapshot and Template Creation

The section detailed the creation of a recovery point snapshot and a VM template from G03WIN11TEMP. The process included the creation of a recovery point named “WIN11Config”. The template allowed consistent and fast deployment of pre-configured virtual machines without reinstalling and reconfiguring the operating system from scratch. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Clicked *Compute* dropdown | *VMs* | *G03WIN11TEMP* | *More* dropdown | *Create Recovery Point*
2. Typed WIN11Config
3. Clicked **Create** in Create Recovery Point
4. Selected **Create VM Template** in *Summary Page*
5. Clicked **Next** in *Create Template from VM* | **Save**

G03S3WIN11VM Deployment

The section detailed the deployment of G03S3WIN11VM from the G03WIN11TEMP template. Deploying from a template ensured a standardized configuration across VMs in the environment and reduced setup time. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section). **Refer to Appendix B.**

1. Clicked *Compute* dropdown | *VMs* | **Create VM from Template** | *VM Template* dropdown | *G03WIN11TEMP* | **Begin**
2. Typed G03S3WIN11VM in *Name*
3. Clicked **Next** in *Deploy VM from Template* | **Deploy** | *G03S3WIN11VM* | *More* dropdown | *Power On* | **Launch Console**
4. Typed [secure password] in *Password*
5. Pressed **Enter**

G03S3WIN11VM Opening

The section detailed the procedure for launching the VM console from the Nutanix Compute dashboard and authenticating with the established administrative credentials.

Consistent access to the VM console was necessary for tasks which involved G03SRV03VM.

The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step

1. Pressed **Ctrl + T** in *Microsoft Edge* browser
2. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked *Compute* dropdown | *VMs* | *G03S3WIN11VM* | *Launch Console*
5. Clicked *Windows Screen*
6. Typed [secure password] in *Password*
7. Pressed **Enter**

IP Address Configuration on G03S3WIN11VM

The section detailed the manual configuration of the IPv4 stack, which included setting static IP to 44.38.3.30 and setting preferred DNS to the virtual DC at 44.38.3.20. The IP configurations were necessary for G03S3WIN11VM to locate the DC and successfully join the AD domain. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03S3WIN11VM desktop was expected to be launched (see G03S3WIN11VM Opening section). **Refer to Appendix B.**

1. Right-clicked **Starts**
2. Clicked *Network Connections* | *Ethernet* | **Edit** in *IP assignment* | *Automatic (DHCP)* dropdown | *Manual*
3. Turned *IPv4* on
4. Typed 44.38.3.30 in *IP address*
5. Typed 255.255.255.0 in *Subnet mask*
6. Typed 44.38.3.1 in *Default gateway*
7. Typed 44.38.3.20 in *Preferred DNS server*
8. Clicked **Save**

Domain and Computer Name Configurations on G03S3WIN11VM

The section detailed the process of renaming the computer and authenticating with AD domain administrator credentials within c242-03-a.group03.c24200.cit.lcl domain, which enrolled the machine to c242-03-a.group03.c24200.cit.lcl domain. Successful AD domain enrollment allowed the client to receive AD domain Group Policy Objects (GPO) and authenticate. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03S3WIN11VM desktop was expected to be launched (see G03S3WIN11VM Opening section).

1. Right-clicked **Start**
2. Clicked *Run*
3. Typed sysdm.cpl in *Open*
4. Clicked **OK** in *Run* | **Change...** |
5. Typed G03S3WIN11VM
6. Checked *Domain*
7. Typed c242-03-a.group03.c24200.cit.lcl
8. Clicked **OK** in *Computer Domain/Name Changes*
9. Typed *Administrator* in *Username*
10. Typed [Domain Administrator password] in *Password*
11. Clicked **OK** in *Windows Security* | **OK** in *Computer Domain/Name Changes* | **Close** in *System Properties* | **Restart Now** in *Microsoft Windows* window

G03SRV03 Domain Controller Demotion

The following section showed how G03SRV03 from an AD DC to a non-Active Directory Domain was demoted. Demotion on G03SRV03 machine was needed for Windows 11 Education installation on the same machine. Unsuccessful DC demotion could cause AD domain conflicts. The following steps were documented on G03SRV03. Windows Server 2022 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 prior to the first step.

1. Clicked *Server Manager* in *Microsoft Windows Taskbar* | *Manage* | *Remove Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Server Selection* window
2. Unchecked *Active Directory Domain Services*
3. Clicked *Remove Features* | *Demote this domain controller*
4. Clicked **Next** in *Credentials* window
5. Checked *Proceed with removal*
6. Clicked **Next** in *Warnings* window
7. Checked *Remove this DNS zone*
8. Clicked **Change...**
9. Typed Administrator in *User name* and [c242-03-a password] in *Password*
10. Clicked **OK** in *Window Security* window
11. Clicked **Next** in *Removal Options* window
12. Typed [new c242-03-a password] in *Password* window and *Confirm password* window
13. Clicked **Next** in *New Administrator Password* | **Demote** in *Demotion* | **Close**

G03SRV03 Partitions Deletion

The following section detailed the deletion of all existing disk partitions on G03SRV03. Clearing partitions prepared the environment for the Windows 11 installation. The following steps were documented on G03SRV03. Once the virtual hardware parameters reached a ready state, the installation of Windows 11 on G03WIN11 could proceed (demonstrated in the next section). Windows Server 2022 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 and UEFI Lexar USB Flash was expected to be plugged in G03SRV03 before the first step was started prior to the first step.

1. Pressed **Windows**
2. Clicked *Power* | *Restart* | **Continue**
3. Pressed **F12**
4. Selected *UEFI Lexar USB Flash Drive* in *UEFI Boot*
5. Pressed **Enter** | **Enter** on
en-us_windows_11_consumer_editions_version_24h2_x64_dvd_1d5fcad3 | **Enter** on
Boot in normal mode
6. Clicked **Next** in *Select language settings* window | **Next** in *Select keyboard settings*
7. Checked *I agree everything will be deleted including files, apps, and settings*
8. Clicked **Next** in *Select setup option* | *I don't have a product key* | *Windows 11 Education* |
Next in *Select Image* | **Accept**
9. Clicked **Next** in *Applicable notices and license terms* | *Custom: Install Microsoft Server Operating System only (advanced)*

10. Clicked [Drive 1/2 partitions] | *Delete* | *OK* in *Microsoft Server Operating System Setup* window
11. Pressed **Shift + F10**
12. Typed list disk; select disk 0; clean
13. Clicked *Close* in *Windows 11 Setup* | **Yes** in *Windows 11 Setup*
14. Repeated steps 3-9
15. Selected *Disk 2 Unallocated Space* | **Next** in *Select location to install Windows 11* | **Install** in *Ready to Install*

Windows 11 on G03WIN11 Installation

The section detailed the installation of Windows 11 (Education) on the G03SRV03 machine. The process included booting from the attached ISO, configuring language and keyboard settings, and establishing the initial administrative credentials for the virtual server. The following steps were documented on G03SRV03. The previous section was expected to have been properly executed.

1. Clicked **Yes** in *Is this the right country or region*
2. Selected **Yes** in *Is this the right keyboard layout or input method* | **Skip** in *Want to add a second keyboard layout* | *I don't have internet* in *Let's connect you to a network*
3. Typed Admin in *Enter your name*
4. Clicked **Next** in *Who's going to use this device*
5. Typed [new admin secure password] in *Enter a password*
6. Inputted [new admin secure password] in *Password confirmation*
7. Typed [answer] in *Security question (1 of 3)*

8. Clicked **Next** in *Now add security questions*
9. Repeated steps 7-8 in *Security question (2 of 3)*
10. Repeated steps 7-8 in *Security question (3 of 3)*
11. Clicked **Accept** in *Choosing privacy settings for your device*

IP Address Configuration on G03WIN11

The section detailed the manual configuration of the IPv4 stack, which included setting static IP to 44.38.3.4 and setting preferred DNS to the parent DC at 44.38.3.2. The IP configurations were necessary for G03WIN11 to locate the DC and successfully join the AD domain. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows 11 prior to the first step. **Refer to Appendix B.**

1. Right-clicked **Starts**
2. Clicked *Network Connections* | *Ethernet* | **Edit** in *IP assignment* | *Automatic (DHCP)* dropdown | *Manual*
3. Turned *IPv4* on
4. Typed 44.38.3.4 in *IP address*
5. Typed 255.255.255.0 in *Subnet mask*
6. Typed 44.38.3.1 in *Default gateway*
7. Typed 44.38.3.2 in *Preferred DNS server*
8. Clicked **Save**

Domain and Computer Name Configurations on G03WIN11

The section detailed the process of renaming the computer and authenticating with AD domain administrator credentials within group03.c24200.cit.lcl domain, which enrolled the machine to group03.c24200.cit.lcl domain. Successful AD domain enrollment allowed the client to receive AD domain Group Policy Objects (GPO) and authenticate. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows 11 prior to the first step.

1. Right-clicked **Start**
2. Clicked *Settings* | *System* | *Remote Desktop* | *On* | **Confirm** in *Remote Desktop Settings* window
3. Pressed **Windows + R**
4. Typed sysdm.cpl in *Open*
5. Clicked **OK** in *Run* | **Change...** |
6. Typed G03WIN11
7. Checked *Domain*
8. Typed group03.c24200.cit.lcl
9. Clicked **OK** in *Computer Name/Domain Changes*
10. Typed *Administrator* in *Username*
11. Typed [Domain Administrator password] in *Password*
12. Clicked **OK** in *Windows Security* | **OK** in *Computer Domain/Name Changes* | **Close** in *System Properties* | **Restart Now** in *Microsoft Windows* window

G03SRV01VM Deployment

The section detailed the deployment of G03SRV01VM from the G03WIN2025TEMP template. Deploying from a template ensured a standardized configuration across VMs in the environment and reduced setup time. The following steps were documented on G03SRV03. Windows 11 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section). **Refer to Appendix B.**

1. Clicked *Compute* dropdown | *VMs* | **Create VM from Template** | *VM Template* dropdown | *G03WIN2025TEMP* | **Begin**
2. Typed G03SRV01VM in *Name*
3. Clicked **Next** in *Deploy VM from Template* | **Deploy** | *G03SRV01VM* | **Update** | **Next** in *Configuration* | *Attach Disk* | *Storage Container* dropdown | *G03VM* | *Bus Type* dropdown | *SATA*
4. Typed 200 in *Capacity* | **Save** in *Attach Disk* | **Save** in *Update VM*
5. Clicked *More* dropdown | *Power On* | **Launch Console** | *Send CtrlAltDel (button icon)* |
6. Typed [secure password] in *Password*
7. Pressed **Enter**

G03SRV01VM Opening

The section detailed the procedure for launching the VM console from the Nutanix Compute dashboard and authenticating with the established administrative credentials. Consistent access to the VM console was necessary for tasks which involved G03SRV03VM. The following steps were documented on G03SRV03. Windows 11 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows 11 prior to the first step.

1. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
2. Pressed **Enter**
3. Clicked *Compute* dropdown | *VMs* | *G03SRV01VM* | *Launch Console*
4. Clicked ***Send CtrlAltDel (button icon)*** |
5. Typed [secure password] in *Password*
6. Pressed **Enter**

IP Address Configuration on G03SRV01VM

The section detailed the process of disabling IPv6 and assigning static IP version 4 (IPv4) address (44.38.2.31) and Domain Name System (DNS) settings to G03SRV01VM. Proper IP configuration allowed the new server to communicate with existing domain controllers and services. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV01VM desktop was expected to be launched (see G03SRV01VM Opening section). **Refer to Appendix B.**

1. Clicked *Local Server* | *IPv4 address assigned by DHCP, IPv6 enabled in Ethernet*
2. Double-clicked *Ethernet*
3. Clicked **Properties**
4. Unchecked *Internet Protocol Version 6 (TCP/IPv6)*
5. Double-clicked *Internet Protocol Version 4 (TCP/IPv4)*
6. Checked *Use the following IP address* and *Use the following DNS server addresses*
7. Typed 44.38.3.31 in *IP address*
8. Typed 255.255.255.0 in *Subnet mask*
9. Typed 44.38.3.1 in *Default gateway*
10. Typed 44.38.3.2 in *Preferred DNS server*
11. Typed 44.2.1.44 in *Alternate DNS server*
12. Clicked **OK** in *Internet Protocol Version 4 (TCP/IPv4) Properties* | **OK** *Ether Properties* | **Close** in *Ethernet Status* | **Close in** *Network Connections*

Domain Configuration on G03SRV01VM

The section detailed the process of enrolling G03SRV01VM from a workgroup to the group03.c24200.cit.lcl domain. The process allowed the server to participate in domain-based security, Group Policy, and centralized management. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV01VM desktop was expected to be launched (see G03SRV01VM Opening section). **Refer to Appendix B.**

1. Clicked *Local Server* | *G03SRV01VM* | **Change...**
2. Typed G03SRV01VM in *Computer Name*
3. Checked *Domain*
4. Typed group03.c24200.cit.lcl
5. Clicked **OK** in *Computer Domain/Name Changes*
6. Typed Group03\Administrator in *Username*
7. Typed [Domain Administrator password] in *Password*
8. Clicked **OK** in *Windows Security* | **OK** in *Computer Domain/Name Changes* | **Close** in *System Properties* | **Restart Now** in *Microsoft Windows* window

Promoting G03SRV01VM to Domain Controller

The section detailed the promotion of G03SRV01VM to an AD DC within the group03.c24200.cit.lcl domain. The objective was completed through navigation within the features of Active Directory server manager, and in ensuring the parent DC administrator credentials used during installation, so the prerequisite checks can run successfully. Domain controller abilities were necessary on server G03SRV01VM, so FSMO roles could later be transferred to the DC. The following procedures were executed within the server manager of G03SRV03VM, from any machine with access to the Prism Central management console. The G03SRV01VM desktop was expected to be launched (see G03SRV01VM Opening section).

1. Clicked *Server Manager* in *Windows Taskbar* | *Manage* | *Add Roles and Features* | **Next** > in *Before You Begin* | **Next** > in *Installation type* | **Next** > in *Server Selection*
2. Checked *Active Directory Domain Service* | *DNS Server*
3. Clicked **Add Features** | **Next** > in *Server Roles* | **Next** > in *Features* | **Next** > in *AD DS* | **Install** in *Confirmation* | **Close** in *Results* | (**Notification icon**) in *Server Manager* | *Promote this server to a domain controller* | **Change...**
4. Typed Group03\Administrator in *User name*
5. Typed [Administrator password from lab 2] in *Password*
6. Clicked **OK** in *Windows Security* | *Select...* | group03.c24200.cit.lcl | **OK** in *Select a domain from the forest* | **Next** > in *Deployment Configuration*
7. Typed [New DSRM password] in *Password*
8. Typed [New DSRM password] in *Confirm password*
9. Clicked **Next** > in *Domain Controller Options* | **Next** > in *DNS Options* | **Next** > in *Additional Options* | **Next** > in *Paths* | **Change...**

10. Typed `Administrator` in *User name*
11. Typed [Administrator password from lab 2] in *Password*
12. Clicked **OK** in *Windows Security* | **Next** in *Preparation Options* | **Next** in *Review Options*
| **Install** in *Prerequisites checks*

FSMO Roles Transfer on G03SRV01

The section detailed the transfer of FSMO roles from G03SRV01 server to G03SRV01VM server. FSMO roles transferred allowed the new virtual DC to take over AD operations, which included PDC Emulator, RID Master, and Infrastructure Master functions. The following steps were documented on G03SRV01. Windows Server 2022 was expected to be booted on G03SRV01 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 prior to the first step.

1. Clicked *Server Manager* in Windows Searchbar | *Tools* | *Active Directory Users and Computers*
2. Right-clicked *Active Directory Users and Computers*
3. Clicked *Change Domain Controller...* | *G03SRV01VM.group03.c24200.cit.lcl* | **OK** in *Change Directory Server*
4. Right-clicked *group03.c24200.cit.lcl*
5. Clicked *Operations Masters...* | **Change...** | **Yes** in *Active Directory Domain Services* | **OK** in *Active Directory Domain Services* |
6. Clicked *PDC* | **Change...** | **Yes** in *Active Directory Domain Services* | **OK** in *Active Directory Domain Services*
7. Clicked *Infrastructure* | **Change...** | **Yes** in *Active Directory Domain Services* | **OK** in *Active Directory Domain Services*

8. Clicked **Close** in *Operations Masters*

9. Typed powershell in *Windows Searchbar*

10. Right-clicked *Powershell*

11. Clicked *Run as administrator*

12. Typed `$Server = Get-ADDomainController -Identity`

`"G03SRV01VM.group03.c24200.cit.lcl";`

`Move-ADDirectoryServerOperationMasterRole -Identity $Server`

`-OperationMasterRole SchemaMaster,DomainNamingMaster`

13. Pressed **Enter**

14. Typed `repadmin /replsummary; repadmin /showrepl` in *PS*

C:\Users\Administrator

15. Pressed **Enter**

G03S1WIN11VM Deployment

The section detailed the deployment of G03S1WIN11VM from the G03WIN11TEMP template. Deploying from a template ensured a standardized configuration across VMs in the environment and reduced setup time. The following steps were documented on G03SRV03. Windows 11 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section). **Refer to Appendix B.**

1. Clicked *Compute* dropdown | *VMs* | **Create VM from Template** | *VM Template* dropdown | *G03WIN11TEMP* | **Begin**
2. Typed G03S1WIN11VM in *Name*
3. Clicked **Next** in *Deploy VM from Template* | **Deploy** | *G03S1WIN11VM* | *More* dropdown | *Power On* | **Launch Console**
4. Typed [secure password] in *Password*
5. Pressed **Enter**

G03S1WIN11VM Opening

The section detailed the procedure for launching the VM console from the Nutanix Compute dashboard and authenticating with the established administrative credentials. Consistent access to the VM console was necessary for tasks which involved G03S1WIN11VM. The following steps were documented on G03SRV03. Windows 11 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows 11 prior to the first step

1. Pressed **Ctrl + T** in *Microsoft Edge* browser
2. Typed `https://44.38.3.12:9440/ui/iam/login` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked *Compute* dropdown | *VMs* | *G03S1WIN11VM* | *Launch Console*
5. Clicked *Windows Screen*
6. Typed [secure password] in *Password*
7. Pressed **Enter**

IP Address Configuration on G03S1WIN11VM

The section detailed the manual configuration of the IPv4 stack, which included setting static IP to 44.38.3.32 and setting preferred DNS to the virtual DC at 44.38.3.31. The IP configurations were necessary for G03S1WIN11VM to locate the DC and successfully join the AD domain. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03S1WIN11VM desktop was expected to be launched (see G03S1WIN11VM Opening section). **Refer to Appendix B.**

1. Right-clicked **Start**
2. Clicked *Network Connections* | *Ethernet* | **Edit** in *IP assignment* | *Automatic (DHCP)* dropdown | *Manual*
3. Turned *IPv4* on
4. Typed 44.38.3.32 in *IP address*
5. Typed 255.255.255.0 in *Subnet mask*
6. Typed 44.38.3.1 in *Default gateway*
7. Typed 44.38.3.31 in *Preferred DNS server*
8. Clicked **Save**

Domain and Computer Name Configurations on G03S1WIN11VM

The section detailed the process of renaming the computer and authenticating with AD domain administrator credentials within group03.c24200.cit.lcl domain, which enrolled the machine to group03.c24200.cit.lcl domain. Successful AD domain enrollment allowed the client to receive AD domain Group Policy Objects (GPO) and authenticate. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03S1WIN11VM desktop was expected to be launched (see G03S1WIN11VM Opening section).

1. Right-clicked **Start**
2. Clicked *Run*
3. Typed sysdm.cpl in *Open*
4. Clicked **OK** in *Run* | **Change...** |
5. Typed G03S1WIN11VM
6. Checked *Domain*
7. Typed group03.c24200.cit.lcl
8. Clicked **OK** in *Computer Domain Name Changes*
9. Typed *Administrator* in *Username*
10. Typed [Domain Administrator password] in *Password*
11. Clicked **OK** in *Windows Security* | **OK** in *Computer Domain/Name Changes* | **Close** in *System Properties* | **Restart Now** in *Microsoft Windows* window

G03SRV01 Domain Controller Demotion

The following section showed how G03SRV01 from an AD DC to a non-Active Directory Domain was demoted. Demotion on the machine was needed for Nutanix CE Acropolis Hypervisor AHV installation on the same machine. Unsuccessful DC demotion could cause AD domain conflicts. The following steps were documented on G03SRV01. Windows Server 2022 was expected to be booted on G03SRV02 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 prior to the first step.

1. Pressed **Windows**
2. Typed `Server Manager`
3. Clicked *Server Manager* | *Manage* | *Remove Roles and Features* | **Next** in *Before You Begin* window | **Next** in *Server Selection* window
4. Unchecked *Active Directory Domain Services*
5. Clicked *Remove Features* | *Demote this domain controller*
6. Clicked **Next** in *Credentials* window
7. Checked *Proceed with removal*
8. Clicked *Next* in *Warnings* window
9. Checked *Remove DNS delegation* in removal options
10. Clicked **Next** in *Removal Options* window
11. Typed [new c242-03-b password] in *Password* window and *Confirm password* window
12. Clicked **Next** in *New Administrator Password* | **Demote** in *Demotion* | **Close**

G03SRV01 Partitions Deletion

The following section detailed the deletion of all existing disk partitions on G03SRV01. Clearing partitions prepared the environment for the Nutanix CE AHV installation. [Drive 0/1/2 partitions] referred to any partitions/volumes which are not Unallocated Space of the drive 0, 1, or 2. The following steps were documented on G03SRV01. Windows Server 2022 was expected to be booted on G03SRV01 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows Server 2022 and UEFI Lexar USB Flash was expected to be plugged in G03SRV01 before the first step was started prior to the first step.

1. Pressed **Windows**
2. Clicked *Power | Restart*
3. Pressed **F12**
4. Clicked *UEFI Lexar USB Flash Drive 9707422273028248* in *One-Time Boot Settings*
5. Pressed **Enter | Enter** on
en-us_windows_server_2022_updated_july_2023_x64_dvd_541692c3 | Enter on Boot in normal mode
6. Clicked **Next** in *Microsoft Server Operating System Setup* window | **Install now** | *I don't have a product key | Windows Server 2022 Standard (Desktop Experience) | Next in Select the operating system you want to install*
7. Checked *I accept the Microsoft Software License Terms. If an organization is licensing it, I am authorized to bind the organization*
8. Clicked **Next** in *Applicable notices and license terms | Custom: Install Microsoft Server Operating System only (advanced)*

9. Clicked [Drive 0/1/2 partitions] | *Delete* | *OK* in *Microsoft Server Operating System Setup* window
10. Repeated step 9 until all [Drive 0/1/2 Unallocated Space]
11. Closed *Microsoft Server Operating System Setup* window | **Yes**
12. Repeated step 11

Nutanix Community Edition Acropolis Hypervisor Installation on G03SRV01

The following section detailed the installation of Nutanix CE AHV on G03SRV01. The hypervisor was configured with a static IP address, subnet mask, gateway, and CVM IP address, which were necessary for G03SRV01 to be integrated into the network. The following steps were documented on G03SRV01. G03SRV01 was expected to be powered off and UEFI Lexar USB Flash Drive was expected to be plugged in before the first step was started. **Refer to Appendix B.**

1. Powered G03SRV01 on
2. Pressed **F12**
3. Selected *UEFI Lexar USB Flash Drive 6164134633453092* in *One-Time Boot Settings*
4. Pressed **Enter** in *nvme0n1*
5. Typed 44.38.3.2 in *Host IP Address*
6. Typed 44.38.3.40 in *CVM IP Address*
7. Typed 255.255.255.0 in *Subnet Mask*
8. Typed 44.38.3.1 in *Gateway*
9. Selected *Next Page*
10. Checked *I accept the end user license agreement.*

11. Selected **Start**
12. Unplugged USB when prompted
13. Pressed **Y** in *Please enter 'Y' to reboot now (Y/N):*
14. Pressed **Enter**

Nutanix Community Edition Acropolis Hypervisor Configuration on G03SRV01

The following section detailed the post-installation configuration of the Nutanix cluster on G03SRV01, which was later renamed into G03NUTXSRV01. Steps included cluster creation, cluster name configuration, DNS configuration, and NTP server configuration. Time synchronization and proper network settings were important. The following steps were documented on G03SRV01. The G03SRV01 was expected to be powered on before the first step was started.

1. Typed `root` in *NTNX-ae56819d-A login*
2. Typed `nutanix/4u` in *Password*
3. Typed `ssh nutanix@192.168.5.2` in *[root@NTNX-ae56819d-A ~]#*
4. Typed `yes` in *Are you sure you want to continue connecting*
5. Typed `nutanix/4u` in *nutanix@192.168.5.2's password*
6. Typed `cluster -s 44.38.3.40 --redundancy_factor=1 create` in *[root@NTNX-ae56819d-A ~]#*
7. Typed `ncli cluster edit-params new-name=G03NUTXSRV01` in *nutanix@NTX-ae56819d-A-CVM:44.38.3.40:~\$*
8. Typed `ncli cluster add-to-name-servers servers=44.2.1.44` in *nutanix@NTX-ae56819d-A-CVM:44.38.3.40:~\$*

9. Typed `ncli cluster add-to-ntp-servers servers=tick.cit.lcl` in
nutanix@NTX-ae56819d-A-CVM:44.38.3.40:~\$
10. Typed `ncli cluster add-to-ntp-servers servers=tock.cit.lcl` in
nutanix@NTX-ae56819d-A-CVM:44.38.3.40:~\$

Disabling IPv6 on G03NUTXSRV01

The section detailed the command-line procedures performed on the CVM, which disabled IP version 6 (IPv6) and modified `sysctl.conf`. To prevent potential communication conflicts, IPv6 was disabled at the hypervisor level. The following steps were documented on G03NUTXSRV01. G03NUTXSRV01 was expected to be powered on before the first step was started.

1. Typed `root` in *NTNX-ae56819d-A login*
2. Typed `nutanix/4u` in *Password*
3. Typed `ssh nutanix@192.168.5.2` in *[root@NTNX-ae56819d-A ~]#*
4. Typed `[new secure password]` in *root@NTNX-ae56819d-A ~'s password*
5. Typed `sudo sysctl -w net.ipv6.conf.all.disable_ipv6=1` in
[nutanix@NTNX-ae56819d-A ~CVM]#
6. Pressed **Enter**
7. Typed `echo -e "net.ipv6.conf.all.disable_ipv6 =
1\nnet.ipv6.conf.default.disable_ipv6 =
1\nnet.ipv6.conf.lo.disable_ipv6 = 1" | sudo tee -a
/etc/sysctl.conf > /dev/null` in *nutanix@NTNX-ae56819d-A ~CVM*
8. Typed `sudo sysctl -p` in *nutanix@NTNX-ae56819d-A ~CVM*

9. Typed `sudo reboot` in *nutanix@NTNX-ae56819d-A ~CVM*
10. Typed `manage_ipv6 unconfigure` in *nutanix@NTNX-ae56819d-A*
~CVM:44.38.3.10
11. Pressed **Enter**
12. Typed `manage_ipv6 unconfigure` in *nutanix@NTNX-ae56819d-A*
~CVM:44.38.3.10
13. Pressed **Enter**
14. Typed `rolling_restart cvms` in *nutanix@NTNX-ae56819d-A ~CVM*
15. Pressed **Enter**

Signing Into G03NUTXSRV01 Prism Element Web Page

The following procedures were executed from the desktop of any machine, considering the machine was connected to the CIT network or the CIT VPN. Each task was completed by following the given prompts and entering the Nutanix Prism Element admin credentials, where the password would be changed. The procedures were executed to manage G03NUTXSRV01 cluster within Nutanix Prism Element, and provide reference for future Prism Element procedure sections. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows 11 prior to the first step.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Typed `https://44.38.3.40:9440` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked **Advanced** on Google security page | *Continue to 44.38.3.10 (unsafe)*

5. Typed `admin` in *username*
6. Typed `nutanix/4u` in *password*
7. Typed `[new secure password]` in *password*
8. Typed `[new secure password]` in *re-type password*

Connecting Cluster Prism Element on G03NUTXSRV01 to Nutanix Prism Central

The section detailed the registration process where the G03NUTXSRV01 cluster was linked to the Nutanix PC instance at IP 44.38.3.12, which made cross-cluster resource management possible. The following steps were documented on G03SRV03. Windows 11 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Pressed **Windows + T**
2. Typed `https://44.38.3.40:9440/console/` in *Microsoft Edge Searchbar*
3. Pressed **Enter**
4. Clicked *Home | Settings | Prism Central Registration | Register or create new | **Connect** |*

Next

5. Typed `44.38.3.12` in *IP Address*
6. Typed `9440` in *Port*
7. Typed `admin` in *Username*
8. Typed `[new secure password]` in *Password*
9. Clicked **Connect**

Raise domain functional level on group03.c24200.cit.lcl

The section detailed the elevation of the operational settings for the group03.c24200.cit.lcl domain to the highest supported version. Raising the functional level was necessary, which enabled advanced security features and administrative capabilities provided by the Windows Server 2025 operating system. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV01VM desktop was expected to be launched (see G03SRV01VM Opening section).

1. Clicked *Server Manager* in *Windows Taskbar*
2. Clicked *Tools | Active Directory Domains and Trusts*
3. Right-clicked *group03.c24200.cit.lcl*
4. Clicked *Raise Domain Functional Level* in *Active Directory Domains and Trusts* | **Raise** in *Raise domain functional level* | **OK** in *Raise forest functional level* window | **OK** in *Raise forest functional level* window
5. Selected **OK** in *group03.c24200.cit.lcl Properties* window

Raise domain functional level on c242-03.a.group03.c24200.cit.lcl

The section detailed the elevation of the operational settings for the c242-03-a.group03.c24200.cit.lcl domain to the highest supported version. Raising the functional level was necessary, which enabled advanced security features and administrative capabilities provided by the Windows Server 2025 operating system. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV03VM desktop was expected to be launched (see G03SRV03VM Opening section).

1. Clicked *Server Manager* in *Windows Taskbar*
2. Clicked *Tools | Active Directory Domains and Trusts*
3. Right-clicked *c242-03-a.group03.c24200.cit.lcl*
4. Clicked *Raise Domain Functional Level* in *Active Directory Domains and Trusts* | **Raise** in *Raise domain functional level* | **OK** in *Raise forest functional level* window | **OK** in *Raise forest functional level* window
5. Selected **OK** in *c242-03-a.group03.c24200.cit.lcl Properties* window

Disabling Memory Overcommit on Windows 11 VMs

The section detailed the modification of virtual memory management settings for both Windows 11 VMs, which were G03SRV01VM and G03SRV03VM. Disabling the overcommit feature was necessary for consistent system performance, which then allowed consistent live migration. The following steps were documented on G03SRV03. Windows 11 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Clicked *Compute | VMs | G03SIWIN11VM | More* dropdown | *Power Off* | **OK** in *Power off selected VM* window | **Update** in *G03SIWIN11VM*
2. Unchecked *Enable Memory Overcommit*
3. Clicked **Save** | *VMs | G03S3WIN11VM | More* dropdown | *Power Off* | **OK** in *Power off selected VM* window | **Update** in *G03SIWIN11VM*
4. Unchecked *Enable Memory Overcommit*
5. Clicked **Save**

iSCSI Data Service IP and Virtual IP Configuration on G03NUTXSRV01

The procedures below were executed within the Nutanix Prism Element console for the G03NUTXSRV01 cluster. IP addresses were generated to configure both the iSCSI and Virtual IPs for the cluster. The steps were necessary for the cluster to become discoverable by Nutanix volumes, and so the cluster could be referenced by the CVM. The following steps were documented on G03SRV03. G03SRV01 machine and G03SRV03 machine were expected to be powered on before the first step was started. Windows 11 was expected to be booted on G03SRV03 machine before the first step was documented; likewise, Administrator account was expected to be logged on Windows 11 prior to the first step. **Refer to Appendix B.**

Windows Server 2022 was expected to be booted before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Pressed **Windows + T**
3. Typed `https://44.38.3.40:9440/console/` in *Microsoft Edge Searchbar*
4. Pressed **Enter**
5. Clicked *Home | Settings | Cluster Details*
6. Typed `44.38.3.42` in *iSCSI Data Services IP*
7. Typed `44.38.3.41` in *Virtual IP*
8. Clicked **Save**

Storage Container Creation on G03NUTXSRV01

The section detailed the creation of a new storage container named “G03VM” within the Prism Element interface. Storage container creation was important for organizing virtual machine data, which provided the necessary environment where the new virtual disks would reside. The following steps were documented on G03SRV03. G03SRV01 machine and G03SRV03 machine were expected to be powered on before the first step was started. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step. **Refer to Appendix B.**

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Pressed **Windows + T**
3. Typed `https://44.38.3.40:9440/console/` in *Microsoft Edge Searchbar*
4. Pressed **Enter**
5. Clicked *Home* dropdown | *Storage* | *Table* | + *Storage Container*
6. Typed G03VM in *Name*
7. Clicked **Save**

Subnet Creation on G03NUTXSRV01 Prism Element

The section detailed how a network subnet was created on the Nutanix PE cluster G03NUTXSRV01 to provide virtual machine (VM) network connectivity, which allowed VMs live migration between clusters possible. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, Purdue email was expected to be logged in Microsoft Edge browser on Windows Server 2022 prior to the first step.

1. Clicked *Microsoft Edge* in *Windows Taskbar*
2. Pressed **Windows + T**
3. Typed `https://44.38.3.40:9440/console/` in *Microsoft Edge Searchbar*
4. Clicked *Home* | *Settings* | *Network Configuration* | **Create Subnet**
5. Typed G03NUTXSRV01 in *Subnet Name*
6. Entered 0 in *VLAN ID*
7. Clicked **Save**

VMs Live Migration across Clusters

The section detailed the transfer of active VMs from the local cluster to a remote destination cluster. Execution of the migration was necessary to balance resource utilization and demonstrate high availability across the enterprise infrastructure. The following steps were documented on G03SRV03. Windows 11 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Clicked *Compute* | *VMs*
2. *Selected G03S1WIN11VM*
3. Clicked *More* dropdown | *Power On* | *More* dropdown | *Migrate Across Clusters* | *Please Select* under *Destination Cluster* | *G03NUTXSRV01* | **Next** in *Destination Cluster*
4. Selected *Destination Subnets* dropdown | *G03NUTXSRV01*
5. Clicked **Next** in *Network* | **Migrate** | *VMs*
6. Selected *G03S3WIN11VM*
7. Repeated steps 3-5

Enabling Memory Overcommit on Windows 11 VMs

The section detailed the activation of the memory overcommit feature for the VMs within the cluster, which were G03SRV01VM and G03SRV03VM. Enabling the overcommit feature was necessary for resource optimization and allowed the VMs to share a common pool of physical memory. The following steps were documented on G03SRV03. Windows 11 was expected to be booted before the first step was documented; likewise, the Prism Central browser console was expected to be launched (see Prism Central Logging section).

1. Clicked *Compute* | *VMs* | *G03SIWIN11VM* | *More* dropdown | *Power Off* | **OK** in *Power off selected VM* window | **Update** in *G03SIWIN11VM*
2. Checked *Enable Memory Overcommit*
3. Clicked **Save** | *VMs* | *G03S3WIN11VM* | *More* dropdown | *Power Off* | **OK** in *Power off selected VM* window | **Update** in *G03SIWIN11VM*
4. Checked *Enable Memory Overcommit*
5. Clicked **Save**

DNS Re-Configuration on G03S3WIN11VM

The section detailed the manual configuration of DNS server addresses for the VMs. Modification of network settings was necessary for reliable communication within the AD forest. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03S3WIN11VM desktop was expected to be launched (see G03S3WIN11VM Opening section). **Refer to Appendix B.**

1. Right-clicked **Starts**
2. Clicked *Network Connections | Ethernet*
3. Selected **Edit** in *IP assignment*
4. Typed 44.38.3.31 in *Preferred DNS server*
5. Clicked **Save**

DNS Re-Configuration on G03SRV03VM

The section detailed the manual configuration of DNS server addresses for the VMs. Modification of network settings was necessary for reliable communication within the AD forest. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV03VM desktop was expected to be launched (see G03SRV03VM Opening section). **Refer to Appendix B.**

1. Clicked *Local Server* | *44.38.3.20*
2. Double-clicked *Ethernet*
3. Clicked **Properties**
4. Double-clicked *Internet Protocol Version 4 (TCP/IPv4)*
5. Typed *44 . 38 . 3 . 31* in *Preferred DNS server*
6. Clicked **OK** in *Internet Protocol Version 4 (TCP/IPv4) Properties* | **OK** *Ether Properties* | **Close** in *Ethernet Status*

Windows 11 Activation for G03S1WIN11VM

The section detailed the activation of Windows 11 on G03S1WIN11VM using Purdue's KMS server, which was kms01.itap.purdue.edu. Activation was required to ensure the operating system was fully licensed and all features remain available in the production environment. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03S1WIN11VM desktop was expected to be launched (see G03S1WIN11VM Opening section).

1. Pressed **Windows**
2. Typed `cmd`
3. Right-clicked *Command Prompt*
4. Clicked *Run as Administrator*
5. Clicked **Yes** in *User Account Control*
6. Typed `slmgr.vbs /skms kms01.itap.purdue.edu`
7. Pressed **Enter**
8. Clicked **OK** in *Windows Script Host* window
9. Typed `slmgr /ipk NW6C2-QMPVW-D7KKK-3GKT6-VCFB2`
10. Pressed **Enter**
11. Clicked **OK** in *Windows Script Host* window
12. Typed `slmgr.vbs /ato`
13. Pressed **Enter**
14. Clicked **OK** in *Windows Script Host* window

Windows 11 Activation for G03S3WIN11VM

The section detailed the activation of Windows 11 on G03S3WIN11VM using Purdue's KMS server, which was kms01.itap.purdue.edu. Activation was required to ensure the operating system was fully licensed and all features remain available in the production environment. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03S3WIN11VM desktop was expected to be launched (see G03S3WIN11VM Opening section).

1. Pressed **Windows**
2. Typed `cmd`
3. Right-clicked *Command Prompt*
4. Clicked *Run as Administrator*
5. Clicked **Yes** in *User Account Control*
6. Typed `slmgr.vbs /skms kms01.itap.purdue.edu`
7. Pressed **Enter**
8. Clicked **OK** in *Windows Script Host* window
9. Typed `slmgr /ipk NW6C2-QMPVW-D7KKK-3GKT6-VCFB2`
10. Pressed **Enter**
11. Clicked **OK** in *Windows Script Host* window
12. Typed `slmgr.vbs /ato`
13. Pressed **Enter**
14. Clicked **OK** in *Windows Script Host* window

Windows Server 2025 Activation for G03SRV01VM

The section detailed the activation of Windows Server 2025 on G03SRV01VM using Purdue's KMS server, which was kms01.itap.purdue.edu. Activation was required to ensure the operating system was fully licensed and all features remain available in the production environment. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV01VM desktop was expected to be launched (see G03SRV01VM Opening section).

1. Pressed **Windows**
2. Typed `cmd`
3. Right-clicked *Command Prompt*
4. Clicked *Run as Administrator*
5. Typed `slmgr.vbs /skms kms01.itap.purdue.edu`
6. Pressed **Enter**
7. Clicked **OK** in *Windows Script Host* window
8. Typed `slmgr /ipk TVRH6-WHNXV-R9WG3-9XRFY-MY832`
9. Pressed **Enter**
10. Clicked **OK** in *Windows Script Host* window
11. Typed `slmgr.vbs /ato`
12. Pressed **Enter**
13. Clicked **OK** in *Windows Script Host* window

Windows Server 2025 Activation for G03SRV03VM

The section detailed the activation of Windows Server 2025 on G03SRV03VM using Purdue's KMS server, which was kms01.itap.purdue.edu. Activation was required to ensure the operating system was fully licensed and all features remain available in the production environment. The following steps were documented on G03SRV03. Windows 11 was expected to be booted on G03SRV03 before the first step was documented; likewise, the G03SRV03VM desktop was expected to be launched (see G03SRV03VM Opening section).

1. Pressed **Windows**
2. Typed `cmd`
3. Right-clicked *Command Prompt*
4. Clicked *Run as Administrator*
5. Typed `slmgr.vbs /skms kms01.itap.purdue.edu`
6. Pressed **Enter**
7. Clicked **OK** in *Windows Script Host* window
8. Typed `slmgr /ipk TVRH6-WHNXV-R9WG3-9XRFY-MY832`
9. Pressed **Enter**
10. Clicked **OK** in *Windows Script Host* window
11. Typed `slmgr.vbs /ato`
12. Pressed **Enter**
13. Clicked **OK** in *Windows Script Host* window

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APPENDIX A: PROBLEM SOLVING

The appendix provides a comprehensive overview of problems faced during the configuration of all systems and virtual networks in the Enterprise Virtualization of the systems. Problems faced during the set up of the systems included being unable to correctly delete all partitions on the Dell Pro Tower Plus server before installing the Nutanix Acropolis Hypervisor. The next problem included being unable to disable Dual Stack on the Nutanix AHV in order to properly connect the cluster to prism central. The final problem faced during the set up of Enterprise Virtualization in the devices was transferring FSMO roles from one domain controller to the VM domain controller.

Problem 1: Deleting Partitions From Dell Pro Tower Plus

Problem Description: After installing windows on a bootable USB drive, the intended windows installation did not show up on the boot menu. Changing the file format was attempted along with using Ventoy. Which was an issue as booting windows from the USB was vital to delete existing partitions on the Dell Pro Tower Plus.

Problem Solutions: One solution to the problem was changing the USB drive format to be correctly configured for the system's boot mode, where reformatting between NTFS or ExFAT could both yield different results. Another solution was ensuring USB boot was enabled in the boot menu and all boot configurations were all properly configured. Another solution to the problem was using Rufus to install windows formatted as NTFS.

Solutions Attempted: One solution attempted for the issue was changing the USB format from NTFS to ExFAT, however, the switch did not work as Windows is more compatible with NTFS. Without tools to convert the USB into a bootable drive like Rufus or Ventoy, downloading the windows file did not work. Another solution attempted was using Rufus

to convert the USB into a bootable drive. By using Rufus with the NTFS file system, the correct partition scheme could be properly used.

Final Solution: The issue was resolved by addressing the issue of the USB not being a bootable drive. Using Rufus, the USB drive was converted into a bootable drive with the correct windows installer. After converting to a bootable drive, the USB was shown in the boot menu and the existing partitions were deleted.

Problem 2: Disabling Dual Stack on Nutanix AHV

Problem Description: During the configuration of Nutanix AHV and the cluster, in order to connect the cluster to prism central, dual stack had to be disabled. In order to disable dual stack, SSHing into the CVM and then typing `manage_ipv6 unconfigure` and `manage_ipv6 disable` was necessary. However, after running the two commands, dual stack was still not disabled.

Problem Solutions: One solution to the problem was modifying the network settings in Prism Element to disable dual stack directly through the web console instead of manually in the Nutanix AHV. Another solution to the problem would be attempting to manually change the values while SSH'd into the CVM, as shown in older versions of the Nutanix AHV. Another solution was ensuring a virtual IP and the VLAN ID had set values inside of Prism Element, as the two values are necessary in the process of creating and connecting a cluster to Prism Central. The final solution to the problem was ensuring the system was rebooted after running the commands `manage_ipv6 unconfigure` and `manage_ipv6 disable`.

Solutions Attempted: One solution attempted was manually disabling dual stack through the Pre-AOS 6.5 steps, which involved editing `/etc/sysctl.conf` by manually setting each

value to 1. However, regardless of the attempt, the dual stack was still not disabled. The next solution attempted was rebooting the CVM after running the `manage_ipv6 unconfigure` and `manage_ipv6 disable` commands.

Final Solution: The issue of disabling dual stack was solved by rebooting the CVM after running the `manage_ipv6 unconfigure` and `manage_ipv6 disable` commands. By rebooting, the changes were fully applied to the network stack and the cluster was able to be created and connected to Prism Central.

Problem 3: Transferring FSMO roles

Problem Description: When FSMO roles were transferred from G03SRV01, the original domain controller, to G03SRV01VM, a problem arose. Without being able to transfer FSMO roles, G03SRV01 was unable to be demoted from domain controller and Nutanix AHV could not be installed on the Dell Pro Tower Plus. Due to the transfer of forest wide roles, the target domain was required to be fully recognized as a qualified domain to ensure the command ran to transfer the roles targeted to the correct domain controller.

Problem Solutions: One solution to the problem was ensuring G03SRV01VM was fully promoted and recognized as a domain controller, ensuring replication and communication. Another solution to the problem was ensuring the administrative user of the G03SRV01VM domain was added to both the schema master and domain naming master groups, ensuring the user had sufficient privileges for the FSMO role changes.

Solutions Attempted: The first solution attempted was ensuring both domain controllers could communicate with each other by pinging each other. The solution included ensuring G03SRV01VM was fully promoted to a domain controller. The next solution attempted

was adding the G03SRV01VM administrator to both the schema master and domain naming master groups in Active Directory.

Final Solution: The issue of transferring the forest-wide FSMO roles was solved by adding the G03SRV01VM administrative user to both the schema master and domain naming master groups. A user must be in both the schema master and domain naming master groups because the roles control forest-wide configurations and require higher privileges compared to typical domain admins in order to prevent modification of the schema or domain structure. By adding the administrative user to both the groups, the transfer of the schema master and domain naming master roles worked.

APPENDIX B: SYSTEM AND NETWORK CONFIGURATION

Appendix B provides a comprehensive overview of all physical and virtual systems configured, network topology, and addressing conventions during the Enterprise Virtualization of Group 3's systems. The following tables outline the hardware/software specifications of each system and records of how each system was organized for management. All virtual machines, clusters, and systems are addressed in the appendix.

Table 1*Cluster and Network Configuration*

System Name	Role	Operating System	IP Address
G03NUTXSRV02	Primary Nutanix Cluster Host	Nutanix AHV	44.38.3.19
G03NUTXSRV01	Primary Nutanix Cluster Host	Nutanix AHV	44.38.3.40
G03Prism	Centralized Management Console	Prism Central (HCI)	44.38.3.12

Table 2*Virtual Machine Configuration*

VM Name	Role	Domain	Operating System	IP Address	vCPUs	RAM
G03SRV03	Domain	c242-03-a.	Windows	44.38.3.20	2	8GiB
VM	Controller	group03.c2	Server			

		4200.cit.lcl	2025			
G03S3WI	Workstatio	c242-03-a.	Windows	44.38.3.30	2	4GiB
N11VM	n	group03.c2	11			
		4200.cit.lcl	Education			
G03SRV01	Domain	group03.c2	Windows	44.38.3.31	2	8GiB
VM	Controller	4200.cit.lcl	Server			
			2025			
G03S1WI	Workstatio	group03.c2	Windows	44.38.3.32	2	4GiB
N11VM	n	4200.cit.lcl	11			
			Education			
G03WIN1	Windows	-	Windows	-	2	4GiB
1TEMP	11		11			
	Template		Education			
G03WIN2	Windows	-	Windows	-	2	8GiB
025TEMP	Server		Server			
	2025		2025			
	Template					

Table 3*Cluster Configuration*

Setting	G03NUTXSRV02	G03NUTXSRV01
Storage Container	G03VM	G03VM
Virtual IP	44.38.3.19	44.38.3.41
iSCSI Data Services IP	44.38.3.11	44.38.3.42
VLAN ID	0	0
Cluster Subnet	44.38.3.0/24	44.38.3.0/24
Host IP	44.38.3.3	44.38.3.2
Subnet Mask	255.255.255.0	255.255.255.0
Gateway	44.38.3.1	44.38.3.1
CVM IP	44.38.3.10	44.38.3.40